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Roll No

EE-6005(1)-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Signal Processing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume any missing data.

1. a) Define continuous and discrete time signals with examples. How these signals can be obtained from each other? 7
b) Draw a basic block of a system. State whether the system are static, linear shift invariant, causal and stable $y(n) = nx(n)$. 7
2. a) Draw the basic structure of a linear constant coefficient difference equation. 7
b) Determine whether each of the following sequences are periodic or not. If periodic determine the fundamental period: 7

i) $x(n) = \sin\left(\frac{6\pi n}{7}\right)$

ii) $x(n) = \sin\left(\frac{n}{8}\right)$

3. a) Define the Z transform, what is the relation between Z plane and S plane? 7

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- b) Determine Z transform and ROC of the following signals

$$x(n) = \left(\frac{1}{2}\right)^n u(-n-1) \quad 7$$

4. a) Write and explain briefly time shifting differentiation and linearity properties of Z transform. 7
 b) Find the Z transform and the ROC of the following sequence: 7

$$x(n) = \left(\frac{-1}{4}\right)^n u(n) - \left(\frac{1}{5}\right)^n u(-n-1)$$

5. a) Discuss any three important properties of DFT with suitable examples. 7
 b) Find the DFT of the sequence $x(n) = (1, 0, 1, 0, 1, 0)$ using DFT-FFT algorithm. 7
 6. a) Enumerate the significant difference between FIR and IIR filter. 7
 b) Convert the analog filter with system function

$$H_a(s) = \frac{s + 0.2}{(s + 0.2)^2 + 1.44} \text{ into a digital IIR filter using impulse invariance method.} \quad 7$$

7. a) Explain Bilinear transformation method for designing digital filter in detail and its advantages. 7
 b) Discuss the types of basic structures for realization of digital filter what is meant by finite word length effects in digital filters. 7
 8. a) What are FIR filter? What are advantage of FIR filter? 7
 b) List the steps involved in design of an FIR filter using Kaiser Window. 7

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